BENNETT UNIVERSITY

PROMPT ENGINEERING

B.Tech 5th Semester

MINI-PROJECT GUIDELINES, MILESTONES & ASSESSMENT

Prompt Engineering — Applied AI Mini-Project

Total Assessment: 30 Marks

Team Size	2–6 students
Milestone 1 Deadline	11 October 2026 — 8 Marks
Milestone 2 Deadline	2 November 2026 — 12 Marks
Milestone 3 Deadline	1 December 2026 — 10 Marks

Student & Faculty Reference Document

1. Introduction

The Mini-Project is a 30-mark applied-learning component of the Prompt Engineering course for B.Tech 5th Semester. The project is intended to move students from prompt concepts and experimentation to the design, testing, evaluation and demonstration of a useful prompt-engineering solution. Teams are expected to identify a meaningful problem or use case, develop prompt strategies and workflows, evaluate outputs systematically, document iterations and demonstrate a working solution.

The prescribed Problem Pool/Bank is retained as a common source of real-world problem statements. For this course, students should reinterpret a selected problem through a Prompt Engineering lens: define the user/task context, identify where generative AI or language-model interaction can add value, design prompts and workflows, establish evaluation criteria, and demonstrate reliable and responsible outputs.

2. Mini-Project Objectives

- Translate a real-world problem into a well-defined AI/prompt-engineering task.
- Design effective prompts using appropriate prompt structures, instructions, context, constraints, examples and output specifications.
- Experiment with prompt variants and document an iterative prompt-development process.
- Use structured evaluation criteria to assess relevance, correctness, consistency, completeness, safety and usefulness of outputs, as appropriate to the project.
- Design prompt workflows that may include multi-step prompting, role/task decomposition, structured outputs, retrieval/context injection or tool-assisted workflows where relevant.
- Demonstrate responsible use of generative AI, including privacy, bias, hallucination, safety, copyright and human-oversight considerations where relevant.
- Communicate the technical design, results, limitations and practical value of the project.
- Ensure meaningful and assessable contribution by every team member.

3. Team Formation and Working Rules

- Each Mini-Project team shall consist of 3–5 students.
- One member may be designated as Team Leader/Coordinator for submission and communication purposes.
- Each team shall work on one project/problem.
- Every team member must have a defined role and contribute to research, prompt design, experimentation, evaluation, implementation, documentation or communication.
- The final presentation/viva-voce must include participation from each team member.
- Teams should maintain evidence of individual contribution, including prompt iterations, experiment logs, evaluation sheets, repository/version history, design work, documentation and presentation material.

4. Project / Problem Selection

Students may select their Mini-Project through either of the following routes.

4.1 Select a Problem from the Prescribed Problem Pool

The complete Problem Pool attached as Appendix A is provided as a student-help resource. Teams may select one problem statement from the pool and develop a Prompt Engineering solution around it. The selected problem

should not be treated as a copy-paste assignment; the team must define its own prompt-engineering task, workflow, evaluation approach and prototype.

- Identify the core user/problem context.
- Define the specific task that a prompt-engineering solution will address.
- Identify inputs, context and expected outputs.
- Design and refine prompts through multiple experiments.
- Compare prompt variants and document the reasoning for changes.
- Define measurable evaluation criteria and test cases.
- Demonstrate the final prompt/workflow with representative examples.

4.2 Propose a Self-Defined Problem

Teams may propose their own suitable problem or use case for Prompt Engineering. The problem should be meaningful, feasible within the project period, supported by evidence or a clear use-case rationale, and approved by the concerned faculty member.

Note: A self-defined project requires faculty approval before the team proceeds with it as the assessed Mini-Project.

4.3 Prompt Engineering Project Principles

Criterion	Expectation
Problem clarity	The project clearly defines the user, task, context and intended outcome.
Prompt quality	Prompts use appropriate instructions, context, constraints, examples and output specifications.
Iteration	The team documents prompt versions, experiments and improvements rather than presenting only a final prompt.
Evaluation	Outputs are evaluated using explicit criteria, test cases, rubrics or comparison methods appropriate to the task.
Reliability & limitations	The team identifies failure cases, ambiguity, hallucination risks or other relevant limitations.
Responsible AI	Relevant privacy, bias, safety, copyright, security and human-oversight considerations are addressed.
Reproducibility	Another person should be able to understand and reproduce the core prompt/workflow and evaluation process.

5. Prompt Engineering Skills to be Demonstrated

Depending on the selected project, students should demonstrate relevant skills from the following areas:

- Task decomposition and problem formulation for language-model-based systems.
- Clear instruction writing and prompt structuring.
- Role/context specification and audience-aware prompting.
- Few-shot and example-based prompting where appropriate.
- Constraint-based prompting and output-format specification.
- Structured outputs such as JSON, tables, schemas or other machine-readable formats where appropriate.

- Prompt chaining and multi-step workflows for complex tasks.
- Prompt comparison, A/B experimentation and iterative refinement.
- Context management and grounding using provided documents/data where appropriate.
- Evaluation design: test sets, rubrics, accuracy/relevance/consistency checks and human evaluation.
- Error analysis and prompt failure diagnosis.
- Hallucination and factuality mitigation strategies appropriate to the use case.
- Responsible and safe prompt design, including privacy and sensitive-data handling.
- Documentation and version control of prompts, workflows and experiments.
- Clear communication of limitations and appropriate human oversight.

6. Suggested Prompt Engineering Project Workflow

Stage	Expected Work
1. Problem Definition	Define the user, problem, context, inputs, desired outputs and success criteria.
2. Prompt Task Design	Translate the problem into one or more prompt-engineering tasks.
3. Baseline Prompt	Create an initial prompt and record representative outputs.
4. Prompt Iteration	Experiment with instructions, context, examples, constraints and workflow structure.
5. Evaluation	Test prompt versions against a defined test set or evaluation rubric.
6. Error Analysis	Identify failure cases, inconsistent outputs, hallucinations, ambiguity or other weaknesses.
7. Refinement	Improve the prompt/workflow based on evidence from testing.
8. Prototype / Integration	Demonstrate the prompt as part of a useful application, workflow or interface where feasible.
9. Responsible-Use Review	Check privacy, safety, bias, copyright, security and human-oversight considerations.
10. Final Demonstration	Present the problem, prompt strategy, evidence, results, limitations and working demonstration.

7. Academic Integrity, Originality and Responsible AI Use

- Students must acknowledge external sources, datasets, models, tools, APIs, documents and references used.
- The selected Problem Pool statement may be the starting point, but the team's prompt design, experiments, evaluation, prototype and documentation must be its own work.
- Prompt outputs must not be presented as factual evidence without appropriate verification when factual accuracy matters.
- Students must not fabricate evaluation results, user feedback, testing evidence or implementation claims.
- Sensitive or personal information should not be entered into AI systems unless its use is explicitly permitted and appropriately protected.
- Teams should document significant AI-assisted development where required by faculty/course instructions and remain responsible for the submitted work.

8. Project Timeline and Milestone Deadlines

The project progresses through three assessed milestones. Teams are expected to work continuously between milestones and maintain evidence of prompt development, experimentation, evaluation and implementation.

Milestone	Project Stage	Working Period	Submission Deadline	Marks
Milestone 1	Problem Identification, Use-Case Definition & Prompt Proposal	Project start → 11 Oct 2026	11 October 2026	8
Milestone 2	Prompt Design, Experimentation, Implementation & Demonstration	10 Oct → 2 Nov 2026	2 November 2026	12
Milestone 3	Final Solution, Evaluation, Documentation & Presentation	3 Nov → 1 Dec 2026	1 December 2026	10

9. Milestone 1 — Problem Identification, Use-Case & Prompt Proposal (8 Marks)

Working period: Project commencement to 9 October 2026.

Component	Marks	Expected Evidence / Content
Project Title / Prompt-Engineering Use Case	1.0	Clear project title and concise description of the AI/prompt-engineering use case.
Problem & User Context	1.0	Problem, users, context, inputs and desired outcome are clearly defined.
Reason for Selection	0.5	Why the problem/use case was selected and why prompt engineering is relevant.
Target Users / Stakeholders	0.5	Users, beneficiaries, customers or stakeholders.
Prompt Engineering Task Definition	1.0	Specific task(s) the prompt/workflow will perform; expected inputs and outputs.
Technology / Model / Tool Approach	0.5	Initial model/tool/interface approach and rationale.
Existing Solutions / Gap	1.0	Comparable solutions or workflows and identified gap/opportunity.
Research / Evidence	1.0	Literature, data, user evidence, domain references or credible sources.
Initial Prompt / Workflow Concept	0.5	Baseline prompt or initial workflow with representative expected output.
Industry / Faculty / Senior Feedback	1.0	Feedback from relevant professional/faculty/senior student

Component	Marks	Expected Evidence / Content
		discussions and how it informs the project.

Deadline: 9 October 2026

10. Milestone 2 — Prompt Design, Experimentation, Implementation & Demonstration (12 Marks)

Working period: 10 October 2026 to 2 November 2026.

Component	Marks	Expected Evidence / Content
Prompt / Solution Design Description and Diagrams	2.0	Prompt architecture/workflow, task decomposition, input-output flow, interface or system diagrams as appropriate.
Running / Partial Build / Implementation	3.0	Working prompt workflow, prototype, application, automation or integration demonstrating meaningful implementation.
Demonstration / Pilot Run / Live Deployment of Modules	2.0	Pilot/testing/demo of available modules with representative inputs and outputs.
Updated Project Git Repository / Source Files Drive Folder	2.0	Updated prompts, code, configuration, test data/instructions and source files with evidence of progress.
BMC / Patent / Research Paper / Technical Report / Synopsis / Abstract	1.0	One appropriate business, research, technical or innovation document adapted to the project.
90–150 Second Short Video / Reel	2.0	Concise visual evidence of prompt experiments, working solution and progress up to Milestone 2.

Deadline: 2 November 2026. Team presentations start in class from 3 November 2026; the schedule will be shared on the LMS.

11. Milestone 3 — Final Solution, Evaluation, Documentation & Presentation (10 Marks)

Working period: 3 November 2026 to 1 December 2026.

Component	Marks	Expected Evidence / Content
Prescribed Final Project Report	2.0	Complete report documenting problem, prompt design, experiments, evaluation, results, limitations and responsible-use considerations.
Organization and Structure of Project Report	1.0	Logical structure, clear prompt/workflow documentation, tables/figures, test evidence, references and consistency.

Component	Marks	Expected Evidence / Content
Presentation PPT/PDF/Creatives	1.0	Clear presentation of problem, prompt strategy, experiments, results, limitations and demo.
Team Presentation Viva-Voce / Contribution by Each Member	2.0	Each member demonstrates understanding of the project and their contribution.
LinkedIn Blog Post	2.0	Public-facing project communication describing the problem, prompt-engineering approach, development and results.
YouTube Video/Reel — 3–5 Minutes	2.0	Complete project journey with final prompt/workflow demonstration and key results.

Deadline: 1 December 2026

12. Recommended Final Project Report Structure

Front Matter: Title page, team details, declaration/certificate if prescribed, acknowledgement, abstract/executive summary and contents.

Chapter 1 — Introduction: Problem context, motivation, objectives, users and project scope.

Chapter 2 — Problem & Domain Analysis: Problem evidence, stakeholders, requirements, domain context and constraints.

Chapter 3 — Existing Solutions & Literature: Comparable solutions, literature, prompt-engineering approaches and gap analysis.

Chapter 4 — Prompt / Solution Design: Task definition, prompt architecture, instructions, context, examples, constraints, output schema and workflow diagrams.

Chapter 5 — Prompt Development & Implementation: Prompt versions, experiments, model/tool configuration, prototype/application implementation and integration.

Chapter 6 — Evaluation & Testing: Test set, evaluation criteria, rubric/metrics, comparative results, error analysis and failure cases.

Chapter 7 — Results & Discussion: Results, interpretation, limitations, reliability and lessons from prompt iterations.

Chapter 8 — Responsible AI / Impact / Business / Research Contribution: Privacy, safety, bias, copyright, human oversight, sustainability/social/business/research value as applicable.

Chapter 9 — Conclusion & Future Scope: Key outcomes and future improvements.

References: All cited sources, tools, models, datasets and external materials in a consistent format.

Appendices: Prompt versions, test cases, evaluation sheets, sample inputs/outputs, screenshots, supporting evidence and additional material.

13. Submission and LMS Instructions

- The official submission mechanism for each milestone will be communicated through the LMS.
- The LMS announcement will be treated as the official communication regarding submission links, forms, file formats and submission timing.
- Students must regularly check the LMS for official instructions.
- Do not rely only on WhatsApp, verbal communication or informal group messages for official submission information.
- Each team should retain copies of all submitted files and evidence.
- Where a Git repository or drive folder is required, ensure that the submitted link remains accessible to the evaluating faculty.

14. Prompt Experimentation & Evaluation Evidence

- Maintain a prompt version history showing meaningful iterations.
- Record representative test inputs and outputs for important experiments.
- Define an evaluation rubric or criteria before making broad claims about improvement.
- Compare baseline and refined prompts where appropriate.
- Document failure cases and explain what was changed in response.
- State model/tool limitations and any assumptions that affect the results.
- Avoid cherry-picking only successful outputs when presenting reliability.

15. Evidence of Individual Contribution

Because contribution by each team member is assessed, teams should maintain continuous evidence of individual work. For Prompt Engineering projects this may include prompt design, test-set creation, evaluation, domain research, coding, workflow design, UI development, documentation, responsible-AI review, presentation and communication.

- Maintain a contribution log.
- Use Git/version history where applicable.
- Retain prompt versions and experiment records.
- Record meaningful stakeholder or faculty discussions.
- Ensure every member can explain the project's prompts, workflow, evaluation and limitations during viva.

16. Student Final Checklist

- ☐ Team has 3–5 members.
- ☐ One project/problem has been selected.
- ☐ If self-defined, faculty approval has been obtained.
- ☐ Problem and user context are clearly defined.
- ☐ Prompt-engineering task, inputs and outputs are documented.
- ☐ Relevant model/tool/technology approach is justified.
- ☐ Initial prompt and workflow have been documented.
- ☐ Prompt iterations and experiments are maintained.
- ☐ Evaluation criteria and test cases are defined.

- ☐ Responsible-AI considerations have been reviewed.
- ☐ Milestone 1 submitted by 9 October 2026.
- ☐ Milestone 2 submitted by 2 November 2026.
- ☐ Milestone 3 submitted by 1 December 2026.
- ☐ Git/source folder and prompt artefacts are updated where required.
- ☐ Final report follows the prescribed structure.
- ☐ Each member is prepared for presentation/viva.
- ☐ LinkedIn Blog and YouTube video are completed.
- ☐ External sources, models, tools, datasets and references are acknowledged.

17. Project Timeline at a Glance

Period	Primary Focus	Key Outputs	Assessment
Project Start – 9 Oct 2026	Discover, define and formulate	Problem/use case, user context, research, initial prompt/workflow	M1 — 8 marks
10 Oct – 2 Nov 2026	Design, experiment and implement	Prompt iterations, evaluation, prototype/workflow, demo, source files, progress video	M2 — 12 marks
3 Nov – 1 Dec 2026	Evaluate, finalize and communicate	Final solution, evaluation, report, presentation, viva, LinkedIn, final video	M3 — 10 marks

TOTAL MINI-PROJECT ASSESSMENT: 30 MARKS

APPENDIX A — STUDENT PROJECT HELP: PROBLEM POOL

The Problem Pool below is provided as a common student-help resource for project selection. Students may select one problem from the pool and develop a Prompt Engineering solution around it, or propose a self-defined problem subject to faculty approval.

How to Use This Problem Pool

1. The pool has three parts. Part C has problems written for this course (coding, education, documents, business). Part S has Smart India Hackathon 2026 statements that can be solved as language-model tasks. Part G has briefs based on open data and guidance from international and government organisations.

2. Select one problem and write its ID (for example C7, S3 or G12) in the Milestone 1 form. A self-defined problem is also allowed with faculty approval.

3. The column "Prompt engineering task" is only a starting point. Your team must define its own task, inputs, outputs, test cases and evaluation criteria.

4. The column "Techniques to use" lists techniques from the course that suit the problem. You may use others, but you must justify your choice and compare prompt versions.

5. Each problem should be scoped so that the prompt or prompt workflow is the main work. A large app with one simple prompt will not score well.

6. For health, legal and finance problems, the solution must give information and guidance only. It must not give a diagnosis, legal advice or investment advice, and it must tell the user when to consult a professional. Do not enter real personal data into any AI tool; use made-up or public data.

7. Using a Smart India Hackathon statement for this mini-project does not require registration for the hackathon.

Part C — Course Problems (coding, education, documents, business)

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
C1	Campus helpdesk assistant	Students asking about university rules, fees, exams and hostel	Student question + official policy documents → correct answer with the source section quoted	Document grounding (RAG), refusal when answer is not in the document, evaluation test set
C2	Lecture notes to study pack	Students revising from long notes or slides	Lecture notes → summary, key terms, practice questions with answers	Prompt chaining, structured output, rubric-based evaluation
C3	Exam question generator with quality check	Faculty and teaching assistants	Syllabus topic + Bloom's level → questions, answer key and marking scheme	Few-shot examples, constraints, second prompt as reviewer
C4	Code review assistant for beginner programmers	First-year programming students	Student code → bugs, style issues and hints without giving the full solution	Role prompting, constraints, structured output, test cases
C5	Bug explanation and fix assistant	Students and junior developers	Error message + code → cause, fix and explanation in simple words	Chain-of-thought, step-by-step output, comparison of prompt versions

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
C6	Unit test generator	Developers	Function or class → unit tests covering normal and edge cases	Few-shot examples, output format control, checking tests by running them
C7	Code translation between languages	Developers moving code, e.g. Java to Python	Source code → equivalent code in the target language with notes on differences	Constraints, chaining (translate, then verify), test-based evaluation
C8	Natural language to SQL	Non-technical staff who need reports from a database	Question in English + table schema → SQL query and a plain explanation	Schema as context, few-shot examples, accuracy check on a query set
C9	Resume feedback against a job description	Students applying for placements	Resume + job description → gap list, rewritten bullet points, match score with reasons	Role prompting, rubric in the prompt, structured output
C10	Mock interview question and feedback tool	Students preparing for placement interviews	Job role + student answer → follow-up question and feedback on the answer	Multi-turn prompt design, rubric-based feedback, consistency testing
C11	Invoice and receipt data extraction	Accounts staff in small businesses	Invoice text or image → JSON with vendor, date, items, tax and total	Structured output (JSON schema), few-shot examples, field-level accuracy
C12	Meeting notes to action items	Teams and student clubs	Meeting transcript → decisions, action items with owner and deadline	Structured output, chaining, checking against the transcript
C13	Customer support reply drafting	Support staff of a small online business	Customer complaint + company policy → polite reply that follows the policy	Document grounding, tone constraints, guardrails, human approval step
C14	Product review analysis	Small sellers on e-commerce platforms	Set of reviews → themes, sentiment per theme and suggested improvements	Classification prompts, structured output, self-consistency
C15	Research paper summariser for students	Students doing a literature review	Research paper → structured summary: problem, method, results, limitations	Chaining for long documents, hallucination checks against the paper
C16	Email triage and reply assistant	Faculty or office staff with heavy inboxes	Email → category, priority and a draft reply	Classification, few-shot examples, structured output
C17	FAQ bot from a product manual	Users of a device or software	User question + manual → answer with page or section reference	Document grounding (RAG), refusal rules, evaluation test set

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
C18	Terms and conditions in plain language	Ordinary users signing up for apps and services	Terms or privacy policy → plain summary and list of risky clauses	Chaining, structured output, accuracy check by human reading
C19	Bilingual explainer (English and Hindi or a regional language)	Learners who study better in their mother tongue	Technical concept → explanation in the chosen language at the chosen level	Audience-aware prompting, few-shot examples, human evaluation
C20	Timetable and study plan generator	Students planning exam preparation	Subjects, exam dates and free hours → day-wise study plan in a table	Constraint-based prompting, structured output, constraint checking
C21	Data analysis report writer	Students and analysts with CSV data	Dataset summary or CSV → findings in plain language with suggested charts	Tool-assisted workflow, chaining, checking numbers against the data
C22	Fact-check assistant for social media claims	General public	Claim + trusted sources → verdict with evidence and confidence level	Document grounding, chain-of-thought, self-consistency
C23	Prompt safety tester (red-team kit)	Teams building chatbots	A system prompt → set of attack prompts, results and suggested fixes	Prompt injection tests, guardrails, evaluation table
C24	Multi-step research agent	Students and analysts	Research question → plan, search, notes and a cited short report	ReAct / tool use, chaining, citation checks

Part S — Smart India Hackathon 2026 Statements Suited to Prompt Engineering

Selected from the official SIH 2026 problem statements. Official catalogue: https://www.sih.gov.in/sih2026PS. Read the full statement on the portal before you finalise.

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
S1	Farmers Face Long Waits & Uncertainty on Procurement Schedules/Status [SIH26032]	Ministry of Consumer Affairs, Food & Public Distribution	Farmer question in a local language + procurement data → status reply and next steps	Document grounding, multilingual prompts, structured output
S2	AI-Powered Vernacular Pedagogy & Real-Time Translation Tool for Mother-Tongue Primary Education [SIH26042]	Government of Jharkhand	English lesson text → explanation and classroom activities in the mother tongue	Audience-aware prompting, few-shot examples, human evaluation

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
S3	Digital Platform to Crowdsource Societal Challenges via University–Industry Collaboration [SIH26043]	Government of Jharkhand	Problem submitted by a citizen → category, short summary and duplicate check	Classification, structured output, consistency testing
S4	Portal for Academia–Industry Collaboration on Skill Mapping, Internships & Placement [SIH26044]	Ministry of Ayush	Student profile + job or internship description → skill gaps and matching roles	Structured output, rubric in the prompt, chaining
S5	IP-SAKTI Sahayak — Multilingual AI Assistant for IP & Regulatory Guidance in Ayurveda [SIH26045]	Ministry of Ayush	Question on IP or regulation in Ayurveda + official documents → answer with source	Document grounding (RAG), multilingual prompts, refusal rules
S6	Patient Case-Taking Software for Ayurveda Clinical Documentation [SIH26047]	Ministry of Ayush	Doctor's free-text notes → structured case record (uses made-up data only)	Structured output (JSON), few-shot examples, privacy rules
S7	WeatherGPT — Conversational AI for Weather Forecasting, Alerts & Climate Information [SIH26068]	Ministry of Earth Sciences	Weather question + forecast data → plain answer and alert message	Context injection, tool use, checking answers against the data
S8	CAPACITY CONNECT — Digital Capacity-Building & Learning-Management Portal [SIH26075]	Ministry of Earth Sciences	Training material → course outline, quiz and learner feedback	Chaining, structured output, rubric-based evaluation
S9	Multilingual Cooperative Governance & Legal-Assistance Chatbot [SIH26088]	Ministry of Cooperation	Question on cooperative rules + legal text → answer in the user's language with source	Document grounding (RAG), multilingual prompts, guardrails
S10	AI-Driven Market Linkage & Smart Cataloguing App for Marginalised Artisans [SIH26090]	Ministry of Social Justice & Empowerment (MoSJE)	Artisan's product details or photo → catalogue title, description and tags	Few-shot examples, multimodal prompts, output format control
S11	AI-Driven Hyper-Local Business Advisory & Financial-Structuring Assistant for Rural Micro-Entrepreneurs [SIH26091]	Ministry of Social Justice & Empowerment	Micro-entrepreneur's business details → simple business advice and cost plan	Role prompting, constraints, human oversight note
S12	AI-Driven Scheme-Matching for Marginalised Entrepreneurs [SIH26092]	Ministry of Social Justice & Empowerment	Entrepreneur profile + scheme documents → matching schemes with eligibility reasons	Document grounding, structured output, accuracy test set
S13	AI-Powered Institutional Archive for Heritage Memorials, Manuscripts & Ambedkar Studies [SIH26096]	Ministry of Social Justice & Empowerment	Question about archive documents → answer with citation to the manuscript	Document grounding (RAG), citation checks

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
S14	AI-Driven Voice Assistant for Livelihood Mapping & Skilling Recommendations (PM-AJAY) [SIH26097]	Ministry of Social Justice & Empowerment	Spoken or typed profile of a worker → livelihood options and training suggestions	Multi-turn prompts, structured output, multilingual prompts
S15	AI-Enabled Learning Platform for Competency-Gap Detection & Auto-Generated Quizzes (iGOT Karmayogi) [SIH26101]	Ministry of Statistics & Programme Implementation	Learner's answers → competency gaps and auto-generated quiz	Chaining, few-shot examples, question quality rubric
S16	Accessibility & Quality of Public Healthcare Services in Rural/Underserved Areas [SIH26133]	Government of Maharashtra	Patient question → which health service to go to and what to carry (no diagnosis)	Guardrails, refusal rules, document grounding
S17	Aligning Skill-Development Programmes with Industry & Emerging Job-Market Requirements [SIH26134]	Government of Maharashtra	Course syllabus + job postings → skill gap report and suggested updates	Chaining, structured output, comparison of prompt versions
S18	AI-Based Interactive Quantum-Algorithm Learning Platform [SIH26140]	Egreen Quanta	Learner question on a quantum algorithm → step-by-step explanation and practice question	Chain-of-thought, audience-aware prompting, correctness checks
S19	AI/NLP Engine to Detect Serious-Injury-and-Fatality Precursors in Safety Incident Reports [SIH26165]	Oil India Limited	Safety incident report → risk category, warning signs and reason	Classification, few-shot examples, accuracy on labelled reports
S20	ORCA — Marine Ecosystem Reasoning with Collaborative (Agentic) AI Agents [SIH26176]	Indian Space Research Organisation (ISRO)	Question on marine data → plan, agent steps and a reasoned answer	ReAct / agents, tool use, chaining
S21	Field-Deployable AI Smart-Farming Assistant for Crop-Disease, Pest & Irrigation Advisory [SIH26180]	Qualcomm Inc	Farmer's description of a crop problem → likely causes and advice with a caution note	Few-shot examples, guardrails, multilingual prompts

Part G — Briefs Based on Open Data and Guidance from Global and Government Organisations

Use the named organisation's public data and guidance as the evidence and as the documents your prompts are grounded in.

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
G1	Precision & Climate-Resilient Agriculture Advisory Platforms	FAO Digital Agriculture Programme — https://www.fao.org/digital-agriculture	Farmer's crop, location and season → advisory message in simple language	Context injection, multilingual prompts, guardrails

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
G2	AI-Assisted Personalised Learning for Underserved Learners	UNESCO — Artificial Intelligence in Education — https://www.unesco.org/en/digital-education/artificial-intelligence	Topic + learner level → personalised explanation and practice questions	Audience-aware prompting, chaining, human evaluation
G3	Community Legal-Aid & Access-to-Justice Digital Platforms	UNDP — Rule of Law, Justice, Security & Human Rights — https://www.undp.org/rule-of-law	Citizen's legal question + legal-aid documents → plain information and where to get help (no legal advice)	Document grounding, refusal rules, guardrails
G4	Open-Government-Data-Driven Welfare-Scheme Discovery Tools	Open Government Data (OGD) Platform India — https://data.gov.in	Citizen profile + scheme documents → eligible schemes with reasons and documents needed	Document grounding (RAG), structured output, accuracy test set
G5	Disaster-Relief Coordination & Situation-Report Digital Tools	UN OCHA ReliefWeb — https://reliefweb.int	Field updates → situation report in a fixed format	Structured output, chaining, checking against the inputs
G6	Renewable-Energy Subsidy & Adoption Navigation Tools	International Renewable Energy Agency (IRENA) — https://www.irena.org	Household details + subsidy documents → applicable subsidy and steps to apply	Document grounding, structured output
G7	Digital Career & Skills-Matching Tools for Informal-Sector Workers	International Labour Organization (ILO) — Future of Work — https://www.ilo.org/global/topics/future-of-work	Worker's skills and experience → matching jobs and a simple resume	Structured output, few-shot examples
G8	Public-Sector AI Transparency & Explainability Dashboards	OECD.AI Policy Observatory — https://oecd.ai	Description of a public AI system → plain-language explanation for citizens	Audience-aware prompting, rubric-based evaluation
G9	Open-Budget Transparency & Citizen-Engagement Platforms	International Budget Partnership — https://internationalbudget.org	Budget document + citizen question → answer with figures and page reference	Document grounding, checking numbers against the source
G10	Assistive Technology for Persons with Disabilities	W3C Web Accessibility Initiative (WAI) — https://www.w3.org/WAI/	Web page or image → alt text, captions and simpler wording	Multimodal prompts, constraints, human evaluation
G11	Multilingual Consumer-Grievance & Service-Access Tools	National Consumer Helpline (INGRAM), Dept. of Consumer Affairs — https://consumerhelpline.gov.in	Consumer complaint in any language → category, complaint draft and next steps	Classification, multilingual prompts, structured output

ID	Problem	Users / Source	Prompt engineering task (input → output)	Techniques to use
G12	Digital Financial-Literacy Tools for Low-Income Households	CGAP (Consultative Group to Assist the Poor), World Bank Group — https://www.cgap.org	Money question from a low-income user → simple explanation (no investment advice)	Audience-aware prompting, guardrails, refusal rules
G13	Disaster-Resilient Housing & Building-Code Advisory Tools	UN Office for Disaster Risk Reduction (UNDRR) — https://www.undrr.org	House details + building-code guidance → safety checklist	Document grounding, structured output
G14	MSME Export-Readiness & Compliance Digital Guidance Tools	International Trade Centre (ITC) SME Trade Academy — https://intracen.org	MSME product and target country → export compliance checklist	Document grounding, chaining, structured output
G15	Heat-Wave Public-Health Advisory Generators	World Meteorological Organization (WMO) — Heat-Health Action — https://public.wmo.int	Heat forecast + audience type → public health advisory message	Context injection, tone and length constraints, few-shot examples
G16	Refugee & Migrant Service-Navigation Digital Tools	UNHCR — The UN Refugee Agency — https://www.unhcr.org	Migrant's question + service documents → which service to use and how, in their language	Document grounding, multilingual prompts
G17	Digital Reading Companions for Early-Grade Literacy	Global Book Alliance — https://globalbookalliance.org	Child's reading level → short story, word meanings and questions	Audience-aware prompting, constraints, safety rules for children
G18	Occupational-Safety Digital Compliance Tools for Construction Workers	ILO — Occupational Safety and Health — https://www.ilo.org/topics/safety-and-health-work	Type of construction work → safety checklist and toolbox-talk script	Document grounding, structured output
G19	Green-Computing & E-Waste Reduction Advisory Tools	UNEP / Global E-waste Monitor — https://www.unep.org	Device details → repair, reuse or recycle advice	Few-shot examples, document grounding

Other Sources for Self-Defined Problems

Teams proposing their own problem may also look at Kaggle, the IndiaAI Innovation Challenge, the Open Government Data (OGD) Platform India, industry and startup challenges, and research papers. Mention the source in the Milestone 1 form. Faculty approval is required.

Disclaimer: The Smart India Hackathon statements and the named organisations are taken from public sources current as of 07 September 2026. Students must check the exact wording on the official source before final adoption.